

MATH 3430 - Week 5

Higher-Order Systems and the Matrix Exponential

1 Overview

In this final week of the semester, we will look at **Higher-Order Systems**, both linear and nonlinear. We will explore one major analytical tool, the **Matrix Exponential** e^A , for the $n \times n$ matrix A .

We will work primarily with 3×3 systems of the form

$$\begin{aligned}x' &= f(x, y, z) \\y' &= g(x, y, z) \\z' &= h(x, y, z)\end{aligned}$$

though the techniques generalize “easily” (in theory if not in actual computational practice) to any $n \times n$ system.

We will also get a glimpse of **chaotic behavior** by looking at one of the most famous examples in all of nonlinear dynamics, the **Lorenz Equations**.

The Take-Home Exam is due on Wednesday, so please be sure to keep that in mind. On Thursday, we will spend some time wrapping the semester up and reviewing for the cumulative final exam, which will be on Friday. The Final Exam **WILL** include material covered this week in class, so please make sure that you take the time to understand what we are doing and ask any questions that you might have.

2 Schedule and Checklist

Here is a brief schedule for the week. Please note that this is subject to change, so you need to adjust to what we do in class, regardless of what it says here. Think of this as a pretty accurate guideline, though.

- (M) Intro to Higher-Order Systems and the Matrix Exponential - Linear Systems
- (T) Higher-Order Systems and the Matrix Exponential (day 2) - Nonlinear Systems
- (W) The Lorenz Equations and Chaos. Next Steps. **Take-Home Exam Due.**
- (R) Review and wrap-up.
- (F) **FINAL EXAM!**

Here is a checklist of things to do this week. This is meant to help you keep track of what you've completed and what you still need to work on.

- ☐ Read these notes and ask any questions that you have about anything.
- ☐ Read the sections from Judson and Lebl.
- ☐ Complete all of the homework exercises.
- ☐ Finish the Take-Home Exam and turn it in on Wednesday.
- ☐ Study for the Final.
- ☐ Prepare formula sheet.

3 Terminology and Notation

- The exponential of an $n \times n$ matrix A , e^A .
- The Matrix Exponential, e^{At} .
- The Fundamental Matrix for a system.

4 Equations and Models

- Coupled Mass-Spring System
- Compartment Models
- Multi-Loop Electrical Circuits
- Linear n th Order Equations

Any n th order linear equation of the form

$$y^{(n)} + p_1 y^{(n-1)} + \cdots + p_{n-1} y' + p_n y = 0$$

can be changed to a system of n first-order equations by using the variables

$$x_1 = y, \quad x_2 = y', \quad x_3 = y'', \quad \dots \quad x_n = y^{(n-1)}$$

and writing the corresponding system as

$$\mathbf{x}'(t) = A\mathbf{x}$$

with

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 \\ -p_n & -p_{n-1} & -p_{n-2} & -p_{n-3} & \cdots & -p_1 \end{pmatrix}$$

- Three Species Food Chain
- The Lorenz Equations

5 Important Formulas (analytical material)

- Let A be an $n \times n$ matrix. Then

$$e^A = \sum_{n=0}^{\infty} \frac{1}{n!} A^n$$

where $A^0 = I_n$, the $n \times n$ identity matrix.

- Let A be an $n \times n$ matrix. Then $\mathbf{x}(t) = e^{At} \mathbf{x}_0$ is a solution to the IVP

$$\begin{aligned} \mathbf{x}'(t) &= A\mathbf{x} \\ \mathbf{x}(0) &= \mathbf{x}_0 \end{aligned}$$

- Let A be a matrix with eigenvalue λ and eigenvector $\mathbf{v}$. Then

$$e^{At} \mathbf{v} = e^{\lambda t} \mathbf{v}$$

- Let $\mathbf{x}_i(t)$, $i = 1, \dots, n$ be linearly independent solutions to the $n \times n$ system $\mathbf{x}'(t) = A\mathbf{x}$. Then the fundamental matrix for the system is given by

$$\mathbf{X}(t) = [\mathbf{x}_1(t) \quad \mathbf{x}_2(t) \quad \cdots \quad \mathbf{x}_n(t)]$$

- The matrix exponential, e^{At} , can be given in terms of the fundamental matrix, $\mathbf{X}(t)$, in the following way:

$$e^{At} = \mathbf{X}(t) \mathbf{X}^{-1}(0)$$

- If the $n \times n$ matrix A can be diagonalized by $A = EDE^{-1}$ where D is an $n \times n$ diagonal matrix, then

$$e^{At} = Ee^{Dt}E^{-1}$$

- Combining the previous two items, we have that

$$\begin{aligned} \mathbf{X}(t) &= Ee^{Dt} \\ \mathbf{X}(0) &= E^{-1} \end{aligned}$$

6 Geometric and Visual Aspects (qualitative material)

- Diagram for a coupled harmonic oscillator, in this case a double spring-mass system. This system leads to system of two second-order differential equations, which in turn becomes a 4×4 system of first-order differential equations.

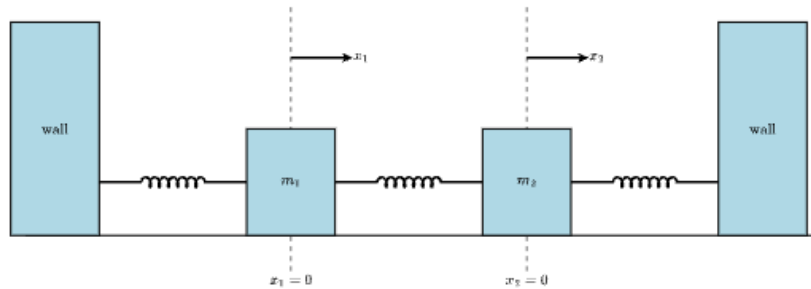


Figure 1: Diagram of a double spring-mass system

- This is a plot of a solution curve to the Lorenz Equations, showing chaotic behavior (the number of times that the curve loops around one “attractor” before jumping to circling around the other “attractor”).

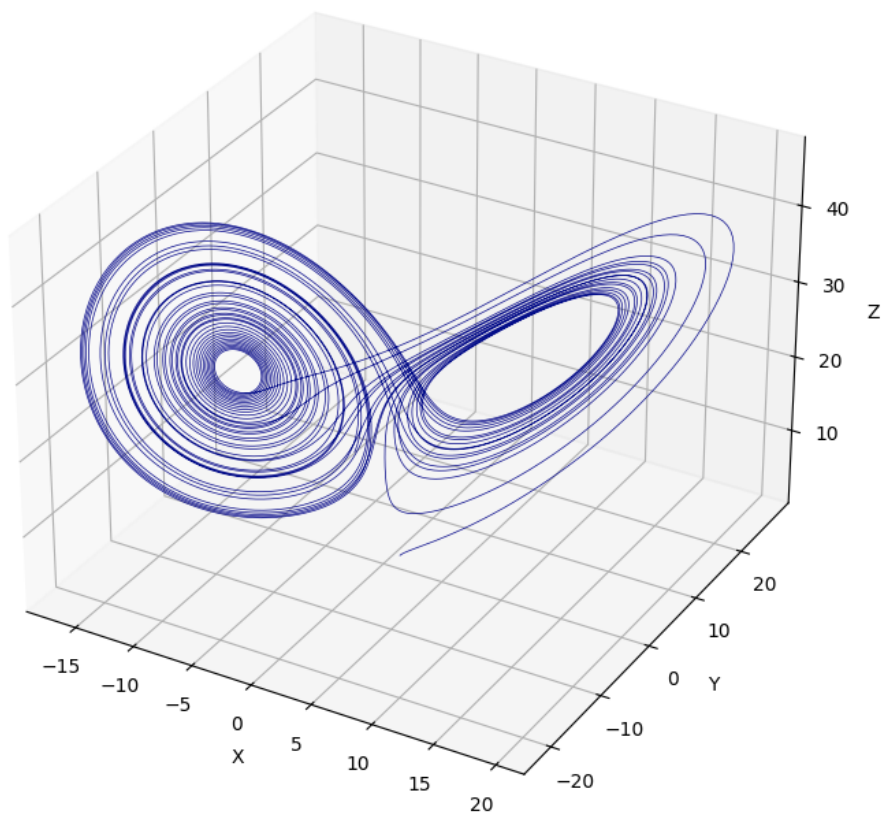


Figure 2: Solution curve for the Lorenz Equations

7 The Big Picture (taken from Strogatz)

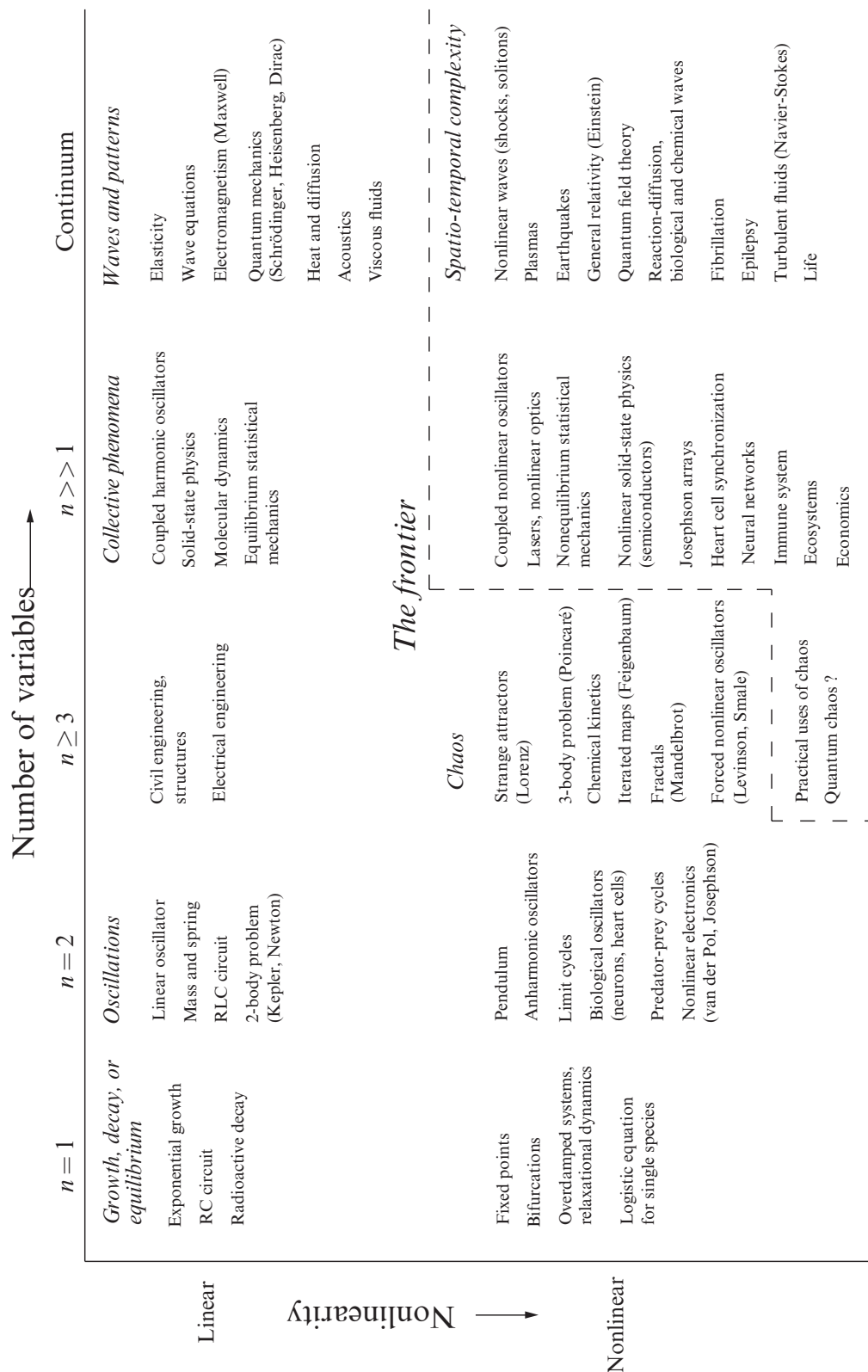


Figure 1.3.1

8 Sage Code (numerical material)

8.1 Commands

- `.exp()`

When this is appended to a matrix A it returns the matrix exponential e^A .

- `%display latex`

This is a formatting command that tells Sage to display all output in the displaymath style of latex. It just makes things nicer to look at and easier to read.

- `identity_matrix(n)`

Creates the $n \times n$ identity matrix.

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8.2 Example Code

- This code generates the matrix exponential e^{tA} directly. It displays the general solution in terms of c_1, c_2, c_3 . The values correspond to Example 3.9.9 in Judson. **PLEASE NOTE** that there is typo in Judson at the end of the example. The eigenvalue is $\lambda = 1$ but in the final line of the example Judson writes it as e^{-t} corresponding to $\lambda = -1$. It should be e^t throughout the example.

```
%display latex
t, c1, c2, c3 = var('t c1 c2 c3')
A = matrix([[-17, -46, -7],[8, 21, 3],[-7, -15, -1]])
x0 = matrix([[c1], [c2], [c3]])
S = (t*A).exp()
S*x0
```

- This code corresponds to the method that Judson uses in example 3.9.9, using the identity $A = \lambda I + (A - \lambda I)$ to give $e^{tA} = e^{t\lambda I} e^{t(A-\lambda I)}$ (this identity is at the beginning of example 3.9.7). This will give the same values as the code in the previous item.

```
%display latex
t, c1, c2, c3 = var('t c1 c2 c3')
A = matrix([[-17, -46, -7],[8, 21, 3],[-7, -15, -1]])
l = A.eigenvalues()[0]
Id = identity_matrix(3)
x0 = matrix([[c1], [c2], [c3]])
S = e^(l*t)*(t*(A-l*Id)).exp()
S*x0
```

- This code generates and plots the solution curves for the three species (tree, moose, wolf) food chain model:

```
t0, m0, w0 = 1, 1, 0.2
x, y, z, t = var('x y z t')
F = [x*(1-x)-x*y, y*(1-y)+x*y - y*z, z*(1-z) + y*z]
```

```

P = desolve_system_rk4(F, [x, y, z], ics=[0, t0, m0, w0], ivar = t, end_points = 10, step
= 0.1)
St = [ [i, j] for i, j, k, l in P]
Sm = [ [i, k] for i, j, k, l in P]
Sw = [ [i, l] for i, j, k, l in P]
p = line(St, color = 'red', legend_label = 'Tree', legend_color = 'red')
p += line(Sm, color = 'green', legend_label = 'Moose', legend_color = 'green')
p += line(Sw, color = 'orange', legend_label = 'Wolf', legend_color = 'orange')
p.show()

```

- The code generates and plots a solution curve for the Lorenz System:

```

s, r, b = 10, 28, 8/3
x, y, z, t = var('x y z t')
F = [s*(y - x), r*x - y - x*z, -b*z + x*y]
P = desolve_system_rk4(F, [x, y, z], ics=[0, 0.1, 0.1, 0.1], ivar = t, end_points = 50,
step = 0.01)
Y = [ [j, k, l] for i, j, k, l in P]
p = list_plot(Y, plotjoined = True)
p.show(axes = True)

```

9 Reading Notes

- Higher-Order Systems (J 3.8) (L 3.6)
- The Matrix Exponential (J 3.9) (L 3.8)
- Chaotic Systems and The Lorenz Equations (L 8.5)

10 Additional Topics

Look in the indices of Judson and Lebl for these topics.

- Strange Attractors
- The Duffing Equation

11 Recommended Books

- Abraham and Shaw
- Blanchard, Devaney, and Hall
- Borrelli and Coleman
- Brannan and Boyce
- Boyce and DiPrima
- Meiss
- Simmons
- Strogatz

12 Homework Exercises

Please do **ALL** exercises. Not all of them will be graded for correctness, but they will all be checked for completion. Failure to do all of them will lower the completion portion of the Homework score.

Please write your solutions up clearly and cleanly, using full English sentences. Try not to have a bunch of arrows pointing from one thing to another or a bunch of things crossed-out and a bunch of other things circled, etc. Presentation matters.

The last two questions are on their own sheet of paper. You can detach this sheet from this packet and do your work/write your answers directly on this sheet of paper. If you do this, please make sure that it is stapled to the other pages that you are turning in. You should only be turning in **ONE** thing for each assignment.

1. Pick one item from the Additional Topics and write a paragraph or two about it. This doesn't have to be extensive, but it should give enough information to explain what the topic is, how it is used, what ideas are involved, etc.